Design Project Proposal

App Name: Automated JobSeeker & CV Generator

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Group : Group B

Problem Identification

Job seekers often face difficulty in finding jobs that match their skills, and in creating different CVs for each job. This process is slow, manual, and stressful.

Key Limitations:

- No automatic skill-based job filtering
- Manual CV generation for each job
- Lack of job notifications and personalized applications

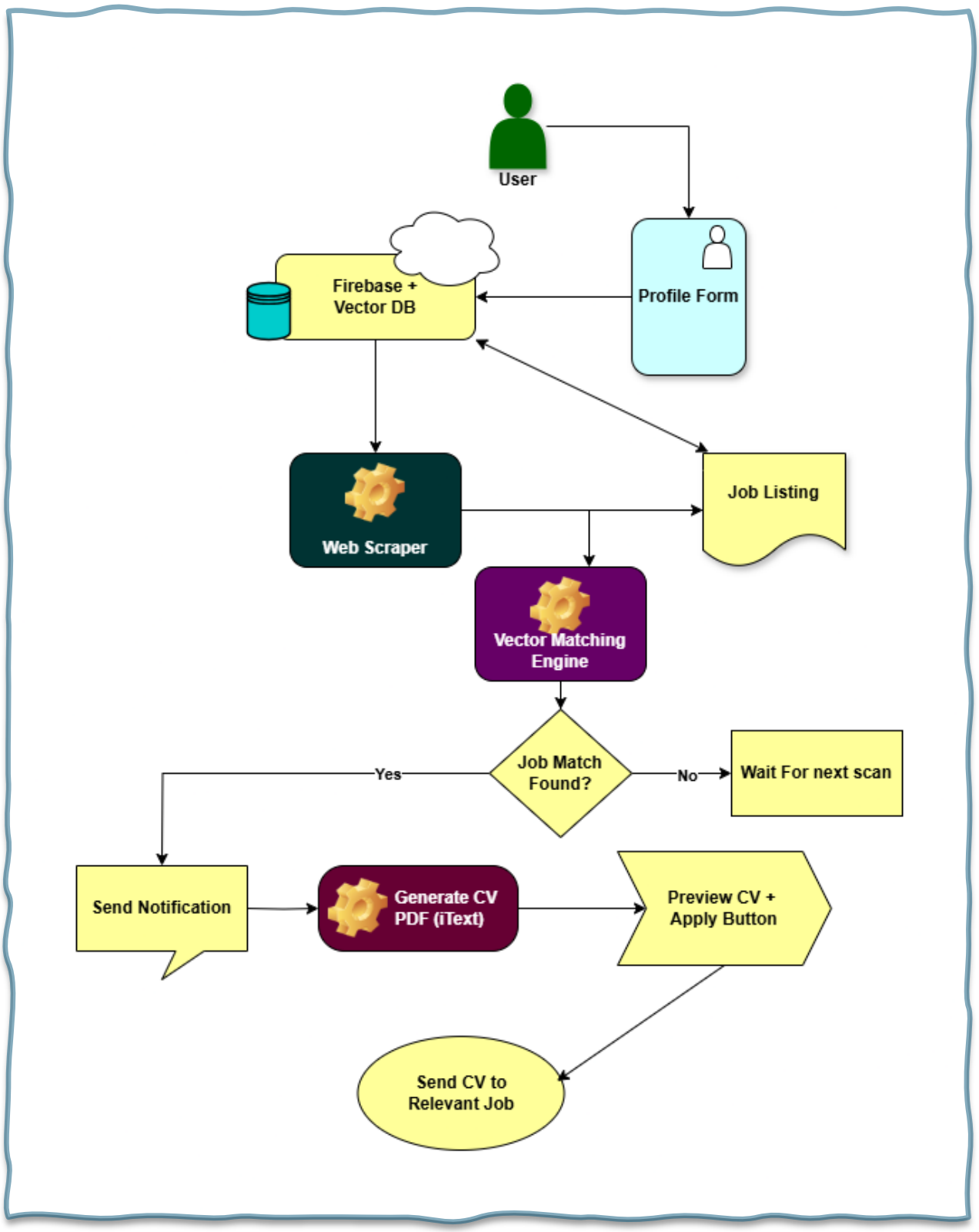
Proposed Approach and Design

This app automates job search and CV creation. The user creates a profile by entering their skills, qualifications, and preferences. After this:

- The app **scrapes** job listings from selected websites
- It checks the user's skills vs job requirements using a **vector database (Neo4j)**
- If a match is found:
    - Notifies the user
    - Generates a tailored CV for that job
    - Allows the user to send the CV with one click

Main Components:

1. Android App (Java + XML using Android Studio)
2. Backend Server for scraping, comparison, and CV generation
3. Firebase for storing user data and sending notifications
4. Vector Database (Neo4j) for skill-job matching
5. iText Java library to generate PDF CVs


User
Firebase +
Vector DB
Profile Form
Web Scraper
Job Listing
Vector Matching
Engine
Job Match
Found?
Yes
No
Wait For next scan
Send Notification
Generate CV
PDF (iText)
Preview CV +
Apply Button
Send CV to
Relevant Job

Expected Outcomes

- Fully automatic job searching and CV generation after user profile creation
- Real-time job match notifications
- One-touch CV submission feature

Advantages:

- Saves time
- Personalized job suggestions
- Reduces manual work

Disadvantages:

- Needs server support for scraping
- Might need updates if job sites change

Key Areas for Applicability

- **Wireframes:** Profile page, job listing page, CV preview page
- **Multimedia:** Profile images, certificate uploads
- **Location Awareness:** Filter jobs by region
- **Sensor Data:** Basic mobile sensors for location
- **Data Model:**
  - Users: ID, Name, Skills, Experience
  - Jobs: Title, Requirements, Link
  - Matches: UserID ↔ JobID
  - CVs: Auto-generated and saved with match ID